

BRIDGE AT CHAINAGE (PADLI GUJARESHWAR)

ABSTRACT OF COST

S.No	Particulars.	Quantity		Rate	Unit	Amount
1	1 Earthwork in excavation for foundation of B-1 structure complete with all lift, dewatering, 1.1.4 shoring and shuttering etc. including refilling of trenches in 250 mm layer, ramming, watering and disposal of surplus soil within a lead of 1000 meters, as per drawing and technical Earth In excavation for foundation of specification. [MORTH Specification: Clause 304]. in ordinary soil [MORTH Specification clause 301] BY MECHANICAL MEANS Depth upto 3m	618.58	Cum	57	P Cum	35259.00
2	Earthwork In excavation for foundation of B-1 structure complete with all lift, dewatering, 1.3.2 shoring and shuttering etc. including refilling of trenches in 250 mm layer, ramming, watering and disposal of surplus soil within a lead of 1000 meters, as per drawing and technical Earth in excavation for foundation of specification. [MORTH Specification Clause 304].In Ordinary/Soft rock (not requiring blasting) [MORTH Specification: Clause 3011 BY MECHANICAL MEANS	572.21	Cum	83	P Cum	47493.02
3	Earthwork in excavation for foundation of B-1 structure complete with all lift, dewatering. 1.4.2 shoring and shuttering etc. including refilling of trenches in 250 mm layer, ramming, watering and disposal of surplus soil within a lead of 1000 meters, as per drawing and technical Earth in excavation for foundation of specification. [MORTH Specification: Clause 304]. In Hard Rock [MORTH Specification : Clause 301, 302, 303]. Blasting Prohibited	1088.31	Cum	670	P Cum	729167.70
4	Structural cement concrete (Design Mix) in open foundations, with form work, at level. Cement concrete for plain/reinforced concrete in open foundations as per drawing and technical specification. [MORTH Specification: Section 1700 & 1500). With 20mm nominal size aggregate P.C.C. M-20 grade	114.44	Cum	4017	P Cum	459709.50
5	Structural cement concrete (Design Mix) B-2 in open foundations, with form work, at 6.6.6 any level. Cement plain/reinforced concrete for concrete in open foundations as per drawing and technical specification. [MORTH Specification : Section 1700 & 1500]. With 20mm nominal size aggregate R.C.C. M-25 grade	961.61	Cum	4341	P Cum	4174341.41

S.No	Particulars.	Quantity		Rate	Unit	Amount
6	Providing and laying steel reinforcement at any level in foundation complete as per 13.1 drawing and technical specification. [MORTH Specification: Section 1600]. HYSD bar reinforcement	64.55	T	88567	P tonne	5716734.15
7	Structural cement concrete (Design Mix) in open foundations, with form work, at level. Cement concrete for plain/reinforced concrete in open foundations as per drawing and technical specification. [MORTH Specification: Section 1700 & 1500). With 20mm nominal size aggregate P.C.C. M-15 grade	272.80	Cum	3038	P Cum	828766.40
8	Structural cement concrete for plain concrete/ reinforced concrete for substructure at any level, complete as per [MORTH Specification: Section 1500 & 1700]. drawing and technical specification. Design mix concrete PCC M-25 Upto 5 mtr. Height	372.09	cum	4319	P Cum	1607050.23
9	Steel reinforcement in substructure HYSD reinforcement	57.16	T	88737	P tonne	5072118.18
10	Providing A.C. pipe Weep holes 150mm B-3 dia in Masonry/plain Concrete/Reinforced 13 Concrete abutment, wing wall/return wall complete as per drawing and technical specification. [MORTH Specification : Clause 2205].	408.50	R mtr	114	P RM	46569.00
11	Providing and and filling in foundation Btrenches, and back filling behind abutments, wing walls, return wall etc., and below pipe bed in layers not exceeding 150mm thick including watering and compacting (IRC-78)	10.11	cum	745	P Cum	7534.19
12	Filter medium behind abutment, wing wall and return wall, complete as per drawing and technical specification. (Same as R7.20.3) [MORTH Specification: Clause 2504].	419.13	cum	560	P Cum	234712.80
13	Structural RCC/PSC upto a height of 5 mtr. for reinforced concrete in superstructure complete including steel formwork, scaffolding etc. As per drawing and technical specification.Design mix concrete. [MORTH Specification : 1500 & 1700].RCC M-30 grade Solid slab superstructure at any level in superstructure complete as per drawing and technical Specification. [MORTH Specification: Section 1600]	297.09	cum	6307	P Cum	1873752.94
14	Steel reinforcement in Superstructure HYSD reinforcement	27.06	T	89654	P tonne	2426395.86

S.No	Particulars.	Quantity		Rate	Unit	Amount
15	Reinforced cement concrete wearing coat B-4 M-30 grade at any level including formwork and reinforcement complete as per drawing and technical Specification. MORTH Specification: Clause 2204, 1500 & 1700].	43.56	Cum	7720	P Cum	336302.50
16	Providing and Laying PCC M-15 grade B-4 leveling course below approach slab 15 complete as per drawing and technical specifications. [MORTH specification: 2700]	14.22	cum	2471	P Cum	35137.62
17	Reinforced cement concrete M-30 grade B-4 approach slab including reinforcement 16 and formwork complete as per drawing and technical specifications. [MoRTH specification: 1500, 1600, 1700, 2704]	21.33	cum	7073	P Cum	150867.09
18	Providing, laying and fixing of Filler Joint Providing and fixing 2mm thick Corrugated Copper Plate in expansion joint complete as per drawing and technical specifications in MORTH specification: clause 2605.	31.88	Sqm	3673	P Sqm	117076.88
				Total	Rs.	18959455.58

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DETAILS OF MEASUREMENTS

S.N.	Particulars	Nos.	Length	Breadth	Height	Qty.	Units
1	Earthwork in excavation for foundation of structure complete with all lift, dewatering, 1.1.4 shoring and shuttering etc. including refilling of trenches in 250 mm layer, ramming, watering and disposal of surplus soil within a lead of 1000 meters, as per drawing and technical Earth In excavation for foundation of specification. [MORTH Specification: Clause 304]. in ordinary soil [MORTH Specification clause 301] BY MECHANICAL MEANS Depth upto 3m						
	Abutment A1	1	x	8.50 x	8.80 x	1.50 =	112.20 Cum
	Piers	1	x	5.90 x	13.10 x	1.65 =	127.53 Cum
		1	x	5.90 x	13.10 x	1.65 =	127.53 Cum
		1	x	5.90 x	13.10 x	1.80 =	139.12 Cum
	Abutment A2	1	x	8.50 x	8.80 x	1.50 =	112.20 Cum
Total Qty.							618.58 Cum
2	Earthwork In excavation for foundation of B-1 structure complete with all lift, dewatering, 1.3.2 shoring and shuttering etc. including refilling of trenches in 250 mm layer, ramming, watering and disposal of surplus soil within a lead of 1000 meters, as per drawing and technical Earth in excavation for foundation of specification. [MORTH Specification Clause 304].In Ordinary/Soft rock (not requiring blasting) [MORTH Specification: Clause 3011 BY MECHANICAL MEANS						
	Abutment A1	1	x	8.50 x	8.80 x	1.50 =	112.20 Cum
	Piers	1	x	5.90 x	13.10 x	1.50 =	115.94 Cum
		1	x	5.90 x	13.10 x	1.50 =	115.94 Cum
		1	x	5.90 x	13.10 x	1.50 =	115.94 Cum
	Abutment A2	1	x	8.50 x	8.80 x	1.50 =	112.20 Cum
Total Qty.							572.21 Cum
3	Earthwork in excavation for foundation of B-1 structure complete with all lift, dewatering, 1.4.2 shoring and shuttering etc. including refilling of trenches in 250 mm layer, ramming, watering and disposal of surplus soil within a lead of 1000 meters, as per drawing and technical Earth in excavation for foundation of specification. [MORTH Specification: Clause 304]. In Hard Rock [MORTH Specification : Clause 301, 302, 303]. Blasting Prohibited						
	Abutment A1	1	x	8.50 x	8.80 x	2.25 =	168.30 Cum
	Piers	1	x	5.90 x	13.10 x	3.00 =	231.87 Cum

S.N.	Particulars	Nos.	Length	Breadth	Height	Qty.	Units
		1 x	5.90 x	13.10 x	3.15 =	243.46	Cum
		1 x	5.90 x	13.10 x	2.85 =	220.28	Cum
	Abutment A2	1 x	8.50 x	8.80 x	3.00 =	224.40	Cum
Total Qty.						1088.31	Cum
4	Structural cement concrete (Design Mix) in open foundations, with form work, at level. Cement concrete for plain/reinforced concrete in open foundations as per drawing and technical specification. [MORTH Specification: Section 1700 & 1500]. With 20mm nominal size aggregate P.C.C. M-20 grade						
	Abutment A1	1 x	8.50 x	8.80 x	0.30 =	22.44	Cum
	Piers	1 x	5.90 x	13.10 x	0.30 =	23.19	Cum
		1 x	5.90 x	13.10 x	0.30 =	23.19	Cum
		1 x	5.90 x	13.10 x	0.30 =	23.19	Cum
	Abutment A2	1 x	8.50 x	8.80 x	0.30 =	22.44	Cum
Total Qty.						114.44	Cum
5	Structural cement concrete (Design Mix) B-2 in open foundations, with form work, at 6.6.6 any level. Cement plain/reinforced concrete for concrete in open foundations as per drawing and technical specification. [MORTH Specification : Section 1700 & 1500]. With 20mm nominal size aggregate R.C.C. M-25 grade						
	Abutment A1	1 x	8.20 x	8.50 x	0.85 =	59.25	Cum
		1 x	4.75 x	8.50 x	1.80 =	72.68	Cum
	Piers	1 x	5.60 x	12.80 x	3.10 =	222.21	Cum
		1 x	5.60 x	12.80 x	3.25 =	232.96	Cum
		1 x	5.60 x	12.80 x	3.30 =	236.54	Cum
	Abutment A2	1 x	8.20 x	8.50 x	0.85 =	59.25	Cum
		1 x	4.75 x	8.50 x	1.95 =	78.73	Cum
Total Qty.						961.61	Cum
6	Providing and laying steel reinforcement at any level in foundation complete as per 13.1 drawing and technical specification. [MORTH Specification: Section 1600]. HYSD bar reinforcement						
	Abutment A1 & A2	2 x	11215		=	22430	Kg
	Piers	3 x	14039		=	42117	Kg
Total Qty.						64547	Kg
						64.55	MT

S.N.	Particulars	Nos.	Length	Breadth	Height	Qty.	Units
7	Structural cement concrete (Design Mix) in open foundations, with form work, at level. Cement concrete for plain/reinforced concrete in open foundations as per drawing and technical specification. [MORTH Specification: Section 1700 & 1500). With 20mm nominal size aggregate P.C.C. M-15 grade						
	Abutment A1	1	x	8.20 x	8.50 x	0.75 =	52.28 Cum
	Piers	1	x	5.60 x	12.80 x	0.75 =	53.76 Cum
		1	x	5.60 x	12.80 x	0.75 =	53.76 Cum
		1	x	5.60 x	12.80 x	0.75 =	53.76 Cum
	Abutment A2	1	x	8.20 x	8.50 x	0.85 =	59.25 Cum
Total Qty.							272.80 Cum
8	Structural cement concrete for plain concrete/ reinforced concrete for substructure at any level, complete as per [MORTH Specification: Section 1500 & 1700]. drawing and technical specification. Design mix concrete PCC M-25 Upto 5 mtr. Height						
	Abutment A1	1	x	8.50 x	1.05 x	5.10 =	45.52 Cum
	Return Wall	2	x	5.50 x	0.75 x	6.50 =	53.63 Cum
	Abut Cap	1	x	8.80 x	0.90 x	0.40 =	3.17 Cum
	Piers	1	x	5.60 x	12.80 x	0.75 =	53.76 Cum
		1	x	5.60 x	12.80 x	0.75 =	53.76 Cum
		1	x	5.60 x	12.80 x	0.75 =	53.76 Cum
	Abutment A2	1	x	8.50 x	1.05 x	5.10 =	45.52 Cum
	Return Wall	2	x	5.50 x	0.75 x	7.25 =	59.81 Cum
	Abut Cap	1	x	8.80 x	0.90 x	0.40 =	3.17 Cum
Total Qty.							372.09 Cum
9	Steel reinforcement in substructure HYSD reinforcement						
	Abutment A1 & A2	2	x	3624		=	7248 Kg
	Piers	3	x	16637		=	49911 Kg
Total Qty.							57159 Kg
							57.16 MT
10	Providing A.C. pipe Weep holes 150mm B-3 dia in Masonry/plain Concrete/Reinforced 13 Concrete abutment, wing wall/return wall complete as per drawing and technical specification. [MORTH Specification : Clause 2205].						
	Abutments	2	x	7	x	5.25	= 73.50 RM
		2	x	7	x	4.50	= 63.00 RM
		2	x	7	x	3.50	= 49.00 RM
		2	x	7	x	2.50	= 35.00 RM
		2	x	7	x	1.50	= 21.00 RM
	Return Walls	4	x	5	x	3.50	= 70.00 RM
		4	x	5	x	2.50	= 50.00 RM
		4	x	5	x	1.50	= 30.00 RM
		4	x	5	x	0.85	= 17.00 RM

S.N.	Particulars	Nos.	Length	Breadth	Height	Qty.	Units
						Total Qty.	408.50 RM
11	Providing and filling in foundation Btrenches, and back filling behind abutments, wing walls, return wall etc., and below pipe bed in layers not exceeding 150mm thick including watering and compacting (IRC-78)						
		1	x	1.80 x	2.00 x	0.30 =	1.08 Cum
		1	x	2.60 x	2.80 x	0.30 =	2.18 Cum
		1	x	2.50 x	2.70 x	0.30 =	2.03 Cum
		1	x	2.10 x	2.30 x	0.30 =	1.45 Cum
		1	x	2.20 x	2.50 x	0.30 =	1.65 Cum
		1	x	2.30 x	2.50 x	0.30 =	1.73 Cum
						Total Qty.	10.11 Cum
12	Filter medium behind abutment, wing wall and return wall, complete as per drawing and technical specification. (Same as R7.20.3) [MORTH Specification: Clause 2504].						
	Abut	2	x	7.90 x	3.10 x	4.50 =	220.41 Cum
	Return	4	x	4.80 x	2.30 x	4.50 =	198.72 Cum
						Total Qty.	419.13 Cum
13	Structural RCC/PSC upto a height of 5 mtr. for reinforced concrete in superstructure complete including steel formwork, scaffolding etc. As per drawing and technical specification.Design mix concrete. [MORTH Specification : 1500 & 1700].RCC M-30 grade Solid slab superstructure at any level in superstructure complete as per drawing and technical Specification. [MORTH Specification: Section 1600]						
	Slab	4	x	10.00 x	8.50 x	0.77 =	261.80 Cum
	Ends	4	x	10.00 x	0.79 x	0.56 =	17.70 Cum
	Kerb	2	x	51.00 x	0.75 x	0.23 =	17.60 Cum
						Total Qty.	297.09 Cum
14	Steel reinforcement in Superstructure HYSD reinforcement AS PER DETAILS ENCLOSED FOR 4 SLABS						
			27064	Kg			
							27.06 MT
15	Reinforced cement concrete wearing coat B-4 M-30 grade at any level including formwork and reinforcement complete as per drawing and technical Specification. MORTH Specification: Clause 2204, 1500 & 1700].						
		1	x	51.25 x	8.50 x	0.10 =	43.56 Cum
						Total Qty.	43.56 Cum
16	Providing and Laying PCC M-15 grade B-4 leveling course below approach slab 15 complete as per drawing and technical specifications. [MORTH specification: 2700]						
		2	x	7.90 x	4.50 x	0.20 =	14.22 Cum
						Total Qty.	14.22 Cum

S.N.	Particulars	Nos.	Length	Breadth	Height	Qty.	Units
17	Reinforced cement concrete M-30 grade approach slab including reinforcement 16 and formwork complete as per drawing and technical specifications. [MoRTH specification: 1500, 1600, 1700, 2704]	2	x	7.90 x	4.50 x	0.30 =	21.33 Cum
						Total Qty.	21.33 Cum
18	Providing, laying and fixing of Filler Joint Providing and fixing 2mm thick Corrugated Copper Plate in expansion joint complete as per drawing and technical specifications in MORTH specification: clause 2605.	5	x	8.50 x	0.75	=	31.88 Sq m
						Total Qty.	31.88 Cum